

In-Course Assessment Examination - II

Department of Physical Science

IT2143 – Visual Programming (Practical)

First Semester

Time: 1 hour 30 minutes

May 2022

Create a folder mentioning your registration number (e.g: 2018ICTXX) and save all your files inside that folder. Follow the guidelines and refer to the demo video of the application.

Suppose you are a .NET developer and you have been requested to develop a “Covid19 Recommender System”, an application to allow a person to enter details in categories such as personal details, vaccination details, symptom details, and safety questions. The entered details should be displayed as shown in sample outputs together with a recommendation to follow.

Use Visual Studio with C# and follow the given instructions below to create this Windows Forms Application.

(Q1) Windows Form Design

Use the controls/tools accordingly and design the windows form as shown following figure. You may use group boxes, labels, text boxes, rich text boxes, numeric up-downs, list boxes, checkboxes, radio buttons, combo boxes, and buttons. Follow a proper naming convention for all controls.

The screenshot shows a Windows Form titled "Covid19 Recommender System". The form is divided into several sections:

- Personal Details:** Includes text boxes for "Name:", "Age:" (with a numeric up-down set to 18 and "Years" label), and "NIC:".
- Vaccination Details:** Includes radio buttons for "Vaccinated" and "Not Vaccinated", and a "Vaccine:" dropdown menu currently showing "No Vaccine".
- Symptom Details:** Includes a list box titled "All Symptoms" containing "Cough", "Difficulty Breathing", "Fever", "Headache", and "Loss of taste or smell". Below it is a text box titled "Symptoms you have".
- Safety Questions:** Includes three checkboxes: "Do you wear masks in public?", "Do you sanitize your hands?", and "Do you follow social distancing?".
- Submit Details:** A button located at the bottom center of the form.

Below the form, there is a large grey rectangular area, likely a placeholder for a recommendation or output.

[10%]

(Q2) Vaccination Details Section

The user should mention whether he/she is vaccinated or not. By default, 'Not Vaccinated' should be selected and the 'Vaccine' dropdown should be disabled with the text 'No Vaccine'. If the user selects 'Vaccinated', then the 'Vaccine' dropdown should be enabled and the 0th index of vaccines should be selected from vaccines Pfizer, Moderna, AstraZeneca, Sinopharm, Sputnik V. Then the user can select a vaccine from the dropdown.

[10%]

(Q3) Symptom Details Section

Under 'All symptoms' all the symptoms which can be seen among Covid19 patients are shown. Such symptoms are Fever, Cough, Tiredness, Loss of taste or smell, Sore throat, Headache, and Difficulty Breathing. The user can select the symptoms he/she is having by double-clicking on a symptom at a time. Then, the selected symptoms should be shown under 'Symptoms you have'. To remove the selected symptoms under 'Symptoms you have', the user should double click on each symptom.

[20%]

(Q4) Safety Questions Section

The user can tick to say 'Yes' and untick to say 'No' to the safety questions such as,

- Do you wear masks in public?
- Do you sanitize your hands?
- Do you follow social distancing?

[20%]

(Q5) 'Submit Details' Button

Initially, the empty output box should have a grey background. When clicking on the "Submit Details" button, all the details entered should be displayed in white text with necessary health recommendations as follows,

1. You are Safe! No action is needed.

While showing this recommendation, the background color should be green with all the following conditions,

- The user is vaccinated.
- The user does not have any symptoms.
- The user has answered 'Yes' to all the safety questions.

2. You may not be Safe! Home quarantine is recommended.

While showing this recommendation, the background color should be orange with the following conditions,

- The user has at least one symptom or a maximum of 3 symptoms.
OR
- The user has answered 'No' for at least one safety question.
OR
- The user is not vaccinated.

3. You may be in Danger! Visit the nearest treatment center.

While showing this recommendation, the background color should be red with the following conditions,

- The user has more than 3 symptoms.

[40%]

Consider the following screenshots.

The screenshot shows the 'Covid19 Recommender System' interface. The 'Personal Details' section includes Name: John Smith, Age: 26 Years, and NIC: 962137543V. The 'Vaccination Details' section shows 'Vaccinated' selected and 'AstraZeneca' as the vaccine. The 'Safety Questions' section has three questions, all of which are checked: 'Do you wear masks in public?', 'Do you sanitize your hands?', and 'Do you follow social distancing?'. The 'Symptom Details' section shows 'All Symptoms' as Cough, Difficulty Breathing, Fever, Headache, and Loss of taste or smell, and 'Symptoms you have' as an empty box. The 'Submit Details' button is visible. The bottom section, titled '***** Safety Questions *****', displays the answers: 'Do you wear masks in public? : Yes', 'Do you sanitize your hands? : Yes', and 'Do you follow social distancing? : Yes'. Below this, the '***** Recommendation *****' section states: 'You are Safe! No action is needed.'

The screenshot shows the 'Covid19 Recommender System' interface. The 'Personal Details' section includes Name: John Smith, Age: 26 Years, and NIC: 962137543V. The 'Vaccination Details' section shows 'Vaccinated' selected and 'AstraZeneca' as the vaccine. The 'Safety Questions' section has three questions, with only 'Do you wear masks in public?' checked. The 'Symptom Details' section shows 'All Symptoms' as Cough, Difficulty Breathing, Fever, Headache, and Loss of taste or smell, and 'Symptoms you have' as Cough. The 'Submit Details' button is visible. The bottom section, titled '***** Safety Questions *****', displays the answers: 'Do you wear masks in public? : Yes', 'Do you sanitize your hands? : No', and 'Do you follow social distancing? : No'. Below this, the '***** Recommendation *****' section states: 'You may not be Safe! Home quarantine is recommended.'

The screenshot shows the 'Covid19 Recommender System' interface. The 'Personal Details' section includes Name: John Smith, Age: 26 Years, and NIC: 962137543V. The 'Vaccination Details' section shows 'Vaccinated' selected and 'AstraZeneca' as the vaccine. The 'Safety Questions' section has three questions, all of which are unchecked: 'Do you wear masks in public?', 'Do you sanitize your hands?', and 'Do you follow social distancing?'. The 'Symptom Details' section shows 'All Symptoms' as Cough, Difficulty Breathing, Fever, Headache, and Loss of taste or smell, and 'Symptoms you have' as Cough and Difficulty Breathing. The 'Submit Details' button is visible. The bottom section, titled '***** Vaccination Details *****', displays the vaccine: 'Vaccine: AstraZeneca'. Below this, the '***** Symptoms *****' section lists: 'Cough' and 'Difficulty Breathing'.

The screenshot shows the 'Covid19 Recommender System' interface. The 'Personal Details' section includes Name: John Smith, Age: 26 Years, and NIC: 962137543V. The 'Vaccination Details' section shows 'Vaccinated' selected and 'AstraZeneca' as the vaccine. The 'Safety Questions' section has three questions, all of which are unchecked: 'Do you wear masks in public?', 'Do you sanitize your hands?', and 'Do you follow social distancing?'. The 'Symptom Details' section shows 'All Symptoms' as Cough, Difficulty Breathing, Fever, Headache, and Loss of taste or smell, and 'Symptoms you have' as Cough, Difficulty Breathing, Fever, Headache, and Loss of taste or smell. The 'Submit Details' button is visible. The bottom section, titled '***** Safety Questions *****', displays the answers: 'Do you wear masks in public? : Yes', 'Do you sanitize your hands? : No', and 'Do you follow social distancing? : No'. Below this, the '***** Recommendation *****' section states: 'You may be in Danger! Visit the nearest treatment center.'

Consider the following sample output.

***** Personal Details *****

Name: John Smith

Age: 26 years old

NIC: 962137543V

***** Vaccination Details *****

Vaccine: Sinopharm

***** Symptoms *****

Cough

Fever

***** Safety Questions *****

Do you wear masks in public? : Yes

Do you sanitize your hands? : Yes

Do you follow social distancing? : No

***** Recommendation *****

You may not be Safe! Home quarantine is recommended.